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标题	Negishi's Reagent Versus Rosenthal's Reagent in the Formation of Zirconacyclopentadienes
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摘要（原文节选）

AbstractZirconacyclopentadienes are versatile precursors for a large number of heteroles, which are accessible by Zr‐element exchange reactions. The vast majority of reports describe their preparation by the use of Negishi′s reagent, which is a species that is formed in situ. The zirconacyclopentadiene is then formed by the addition of one equivalent of a diyne or two equivalents of a monoyne moiety to this Negishi species. Another route involves Rosenthal′s reagent (Cp2Zr(py)Me3SiC≡CSiMe3), which then reacts with ⋯

数据表 · Table 1（14 行，含数值行 12 行）

Summary of 1H NMR signals of the Cp rings in C6D6 and the highest conversion [%] for the zirconacyclopentadienes 11 a – l synthesized by using the Rosenthal or Negishi reagents.

Entry	Product	1H δ Cp	Rosenthal	Negishi				
		[ppm]	Max. conversion [%][a] in toluene	t [min]	Remaining yield after 22 h [%]	Max. conversion [%][a]	t [min]	Remaining yield after 22 h [%]
1	11 a	6.00	>99	10	>99	85	10	0
2	11 b	6.37	97	10	97	56	180	0
3	11 c	6.05	98	10	96	15	30	0
4	11 d	5.84	91 (>99 % in THF)	10	88 (63 % in THF)	28	30	0
5	11 e	5.85	>99	10	>99	0 (41 % with BTMSA)	– (60 min)	0
6	11 f	6.00	93	10	93	83	10	0
7	11 g	5.93	91	180[b]	–	88	180[c]	–
8	11 h	5.70	>99 (>99 % in THF)	10	>99 (83 % in THF)	98	10	98
9	11 i	6.09	97	10	90	94	10	67
10	11 j	5.94	96	10	96	88	10	65
11	11 k	6.01	>99	10	>99	97	10	70
12	11 l	6.35	95	10	83	80	30	26

脚注（逐字）：[a] The conversion was calculated based on the consumption of the starting material. [b] Yield obtained after complete isolation of the product after 3 h. [c] Yield obtained after complete isolation of the product after 3 h.

正文对应句（逐字）：Another possibility for the formation of zirconacyclopentadienes is the use of Rosenthal′s reagent 9 (Figure 1).15, 18 The synthesis of Rosenthal's zirconocene is based on the substitution of THF by pyridine of the complex Cp2Zr(THF)Me3SiC≡CSiMe3. Rosenthal and co‐workers exchanged THF with pyridine

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